

Gaussian Mixture Models

Contents

1. Overview	2
2. One normal curve is not enough	3
3. The mixture density	4
3.1 The formula	5
3.2 Why the weights must add up to 1	6
4. Generating data in two steps	6
4.1 The two steps	6
4.2 The latent variable	6
5. Mixtures in two dimensions	7
5.1 The multivariate normal	8
5.2 The mixture in D dimensions	9
6. Responsibilities	9
6.1 The formula	9
6.2 Soft assignment	10
7. Why maximum likelihood has no closed form	11
7.1 The log-likelihood	11
7.2 Setting the derivative for a mean to 0	12
7.3 The circular dependence	13
8. When maximum likelihood breaks	14
9. Mixtures, KDE and k-means	15
9.1 Compared with KDE	15
9.2 Compared with k-means	15
10. Summary	18
11. Sources	18
12. Key terms	18

Where this fits: Pipeline map steps 0 (Foundations), 8 (Model). See the [Course map](#).



- **Builds on:** Covariance and covariance matrix ([Note 48](#)); Normal distribution ([Note 240](#)); Bayes' theorem ([Note 341](#)); Maximum likelihood estimation (MLE) ([Note 631](#)); Expectation maximization (EM) ([Note 641](#)).
- **Compare with:** K-means ([Note 130](#)); Kernel density estimation (KDE) ([Note 253](#)).

1. Overview

Key point: A Gaussian mixture model describes data with several peaks as a weighted sum of a few normal curves. Each point gets a responsibility for every curve: the probability that this curve produced it. Maximum likelihood for a mixture has no closed-form answer.

This Note follows *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020; MML below), Chapter 11, Sections 11.1 and 11.2, with the generative view and the responsibilities as posteriors from Sections 11.4.1 to 11.4.3.

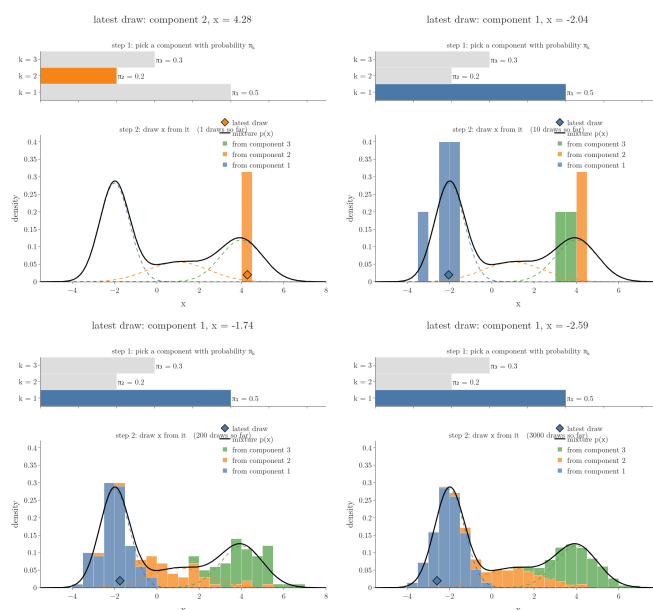


Figure 1: Sampling from a mixture in two steps. Top: pick one of three normal curves with probabilities 0.5, 0.2 and 0.3. Bottom: draw a value from the picked curve. As the draws pile up, their histogram takes the shape of the mixture density (black)

The [maximum likelihood estimation Note](#) fitted one distribution to data. Figure 1 shows data that no single named distribution fits: it has three bumps. A mixture builds such a

shape from simple parts.

This Note covers:

- why one normal curve is not enough (Section 2);
- the mixture density and its weights (Section 3);
- the two-step story that generates mixture data (Section 4, Figure 1);
- mixtures in two dimensions (Section 5);
- responsibilities, computed with Bayes' theorem (Section 6);
- why maximum likelihood has no closed form here (Section 7), and how it can break (Section 8);
- how a mixture compares with KDE and with k-means (Section 9).

How to fit a mixture, the EM algorithm, is the subject of the [expectation maximization Note](#).

2. One normal curve is not enough

Key point: The Old Faithful geyser waits either about 55 or about 80 minutes between eruptions. The best single normal curve sits in the empty gap between the two; a mixture of two normals follows both bumps and has a much higher likelihood.

Think of two bus routes stopping at the same bus stop: one bus comes every 55 minutes or so, the other every 80. Averaging all the waits gives about 71 minutes, a wait that hardly ever happens. One bell curve makes the same mistake.

Figure 2 shows the real case: 272 **observations** (G-1374; records, one row each of the data table) of the waiting time between eruptions of the Old Faithful geyser in Yellowstone (seaborn's `geyser` dataset, saved in `data/old_faithful.csv`). The histogram has two bumps. The red curve is the single normal fitted by maximum likelihood, with the mean 70.9 and standard deviation 13.6 of the data (the [MLE for common distributions Note](#)). Its peak sits near 71 minutes, in the valley between the bumps.

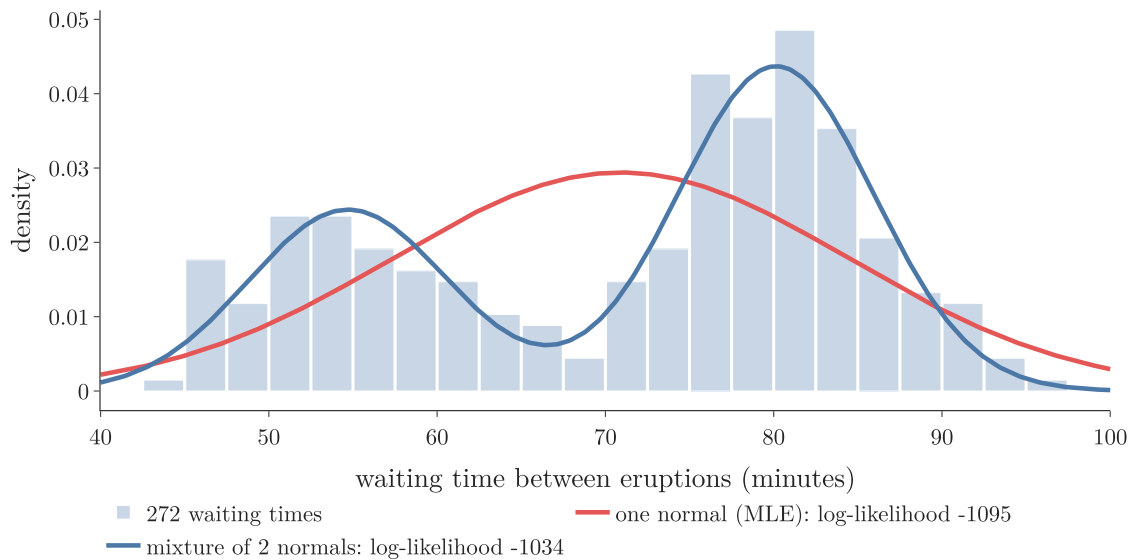


Figure 2: Waiting times between Old Faithful eruptions: the best single normal (red) and a fitted mixture of two normals (blue)

The blue curve is a mixture of two normals: weight 0.36 around 54.7 minutes and weight 0.64 around 80.1 minutes, each with a standard deviation of about 6 minutes. The likelihoods measure the difference:

Model	Log-likelihood of the 272 waiting times
one normal (2 parameters)	-1095
mixture of two normals (5 parameters)	-1034

A higher log-likelihood means the model finds the observed data more probable (the [maximum likelihood estimation Note](#)). The gap of 61 means the data is e^{61} times more probable under the mixture.

MML (Chapter 11 introduction) makes the same point: a single Gaussian has limited modelling power, while a mixture can describe data with several clusters (**multimodal** data, G-1275, see the [measures of central tendency Note](#)).

3. The mixture density

Key point: $p(x) = \sum_k \pi_k N(x \mid \mu_k, \sigma_k^2)$: a weighted sum of K normal densities, with weights that are non-negative and add up to 1.

3.1 The formula

Key point: Each component is a normal curve; each weight says how much of the data it covers.

1. **In words:** a **Gaussian mixture model** (GMM, G-829) adds up K normal densities, called **components**, each multiplied by a **mixture weight** π_k (G-1238). The weights lie between 0 and 1 and add up to 1, so the total area stays 1.
2. **Formula** (MML equations 11.3 and 11.4):

$$p(x | \theta) = \sum_{k=1}^K \pi_k N(x | \mu_k, \sigma_k^2), \quad 0 \leq \pi_k \leq 1, \quad \sum_{k=1}^K \pi_k = 1$$

The parameters are all the weights, means and variances: $\theta = \{\pi_k, \mu_k, \sigma_k^2 : k = 1, \dots, K\}$. Here $N(x | \mu, \sigma^2)$ is the normal PDF of the [normal distribution Note](#), and the second number is the **variance**.

3. **Example:** the mixture of the book's Figure 11.2,

$$p(x) = 0.5 N(x | -2, 0.5) + 0.2 N(x | 1, 2) + 0.3 N(x | 4, 1)$$

At $x = 0$ the three weighted components are 0.0052, 0.0439 and 0.00004, so $p(0) = 0.049$.

Figure 3 draws the three weighted components (dashed) and their sum (black). Where components overlap, their heights add. With $K = 1$ the formula is a single normal curve.

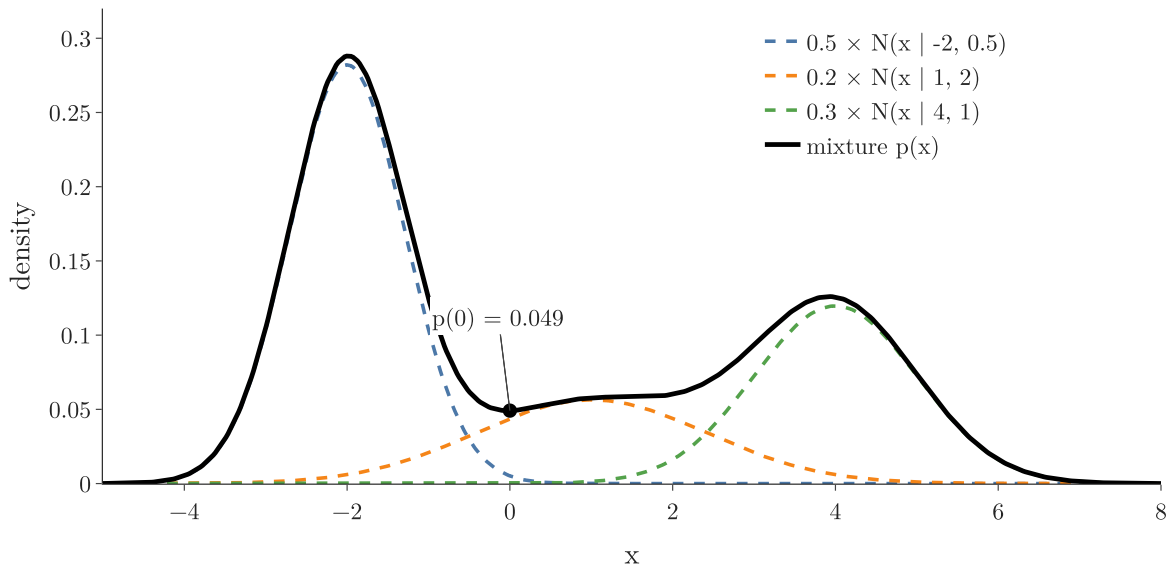


Figure 3: The mixture $0.5 N(-2, 0.5) + 0.2 N(1, 2) + 0.3 N(4, 1)$: weighted components (dashed) and the mixture density (black)

3.2 Why the weights must add up to 1

Key point: Each component has area 1; weights adding up to 1 make the total area 1.

The area under each normal curve is 1 (the [PDF and continuous CDF Note](#)). The area under $\pi_k N(x \mid \mu_k, \sigma_k^2)$ is therefore π_k , and the area under the sum is $\pi_1 + \dots + \pi_K = 1$. In the example, $0.5 + 0.2 + 0.3 = 1$. A sum of this kind, with non-negative weights adding up to 1, is called a **convex combination** (G-475).

4. Generating data in two steps

Key point: To draw from a mixture: first pick a component k with probability π_k , then draw x from that component's normal curve. The picked component is a hidden (latent) variable.

4.1 The two steps

Key point: Step 1 is a weighted die roll; step 2 is an ordinary normal draw.

MML (§11.4.1) describes the **generative process** (G-842) of a mixture:

1. **Pick a component.** Roll a weighted die with K faces: face k comes up with probability π_k .
2. **Draw the value.** Draw x from $N(\mu_k, \sigma_k^2)$, the component that came up.

Repeat for every observation, then forget which component each one came from. Figure 1 runs this for the mixture of Figure 3. The top bar shows the die roll, the diamond the drawn value. After 3,000 draws the histogram matches the mixture density: about half the draws come from the first component, because $\pi_1 = 0.5$ (the Notebook counts 51%).

4.2 The latent variable

Key point: The component that produced a point is a random variable we never observe: a latent variable.

For each observation, write z for the number of the component that produced it. Its distribution is the weights: $P(z = k) = \pi_k$ (MML equation 11.60). Given $z = k$, the value is normal: $p(x \mid z = k) = N(x \mid \mu_k, \sigma_k^2)$.

We only ever see x , never z . A variable that is part of the model but never observed is a **latent variable** (G-1050; Latin *latere*, to lie hidden). Figure 4 shows what is lost. The top row is 300 draws coloured by the component that produced them; the bottom row is the same draws as we receive them. Between the bumps, a grey point could have come from either neighbour, and nothing in the value says which.

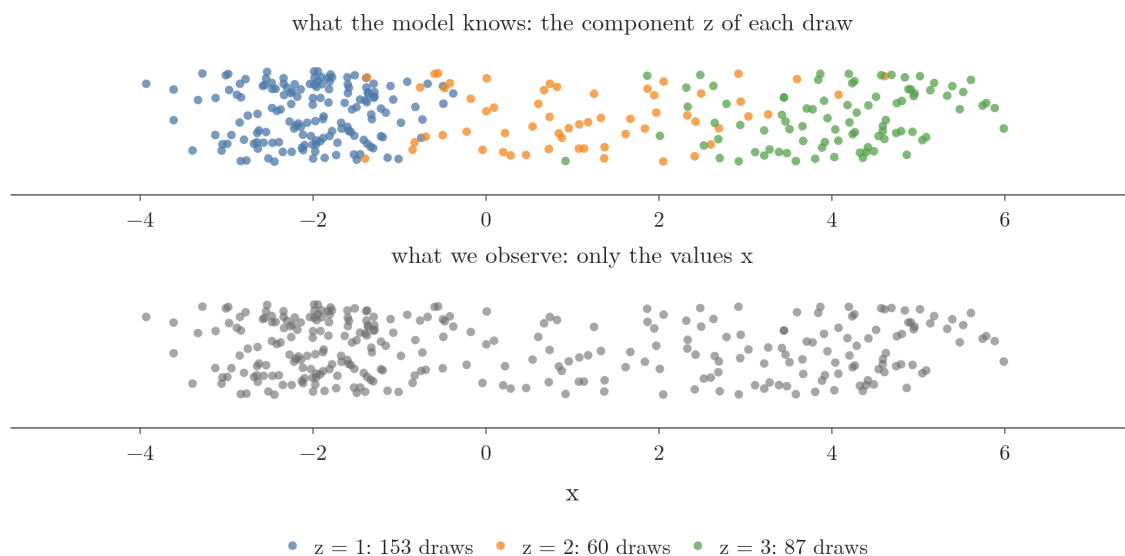


Figure 4: 300 draws from the mixture of Figure 3. Top: coloured by the component z that produced each draw, which the model knows. Bottom: the same draws as we observe them, with z lost.

Adding up over the possible values of z gives back the mixture density (the book's equations 11.65 and 11.66):

1. **In words:** the density of x is the probability of each component times the density of x under it, added over all components.
2. **Formula:**

$$p(x) = \sum_{k=1}^K P(z = k) p(x \mid z = k) = \sum_{k=1}^K \pi_k N(x \mid \mu_k, \sigma_k^2)$$

3. **Example:** at $x = 0$: $0.5 \times 0.0103 + 0.2 \times 0.2197 + 0.3 \times 0.0001 = 0.049$, the same $p(0)$ as Section 3.1.

The sum is the **law of total probability** (G-1053) of the [Bayes problem Note](#), with machines replaced by components.

5. Mixtures in two dimensions

Key point: In two or more dimensions, each component is a multivariate normal with a mean vector and a covariance matrix. Its contour lines are ellipses, which can be stretched and tilted.

5.1 The multivariate normal

Key point: The mean vector sets the centre; the covariance matrix sets the width in each direction and the tilt.

For data with D **features** (G-772; input variables, one column each of the data table), a component is a **multivariate normal distribution** (G-1283) $N(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$. The mean $\boldsymbol{\mu}$ is a vector with one entry per feature. The **covariance matrix** (G-495) $\boldsymbol{\Sigma}$ holds the variances on its diagonal and the covariances off it (see the [covariance and correlation Note](#) and the [PCA step by step Note](#)).

1. **In words:** the density is highest at the mean and falls off with the squared distance from it, measured in a way that accounts for the spread and tilt that $\boldsymbol{\Sigma}$ describes.
2. **Formula** (MML §6.5, equation 6.63):

$$N(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-D/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Here $|\boldsymbol{\Sigma}|$ is the determinant and $\boldsymbol{\Sigma}^{-1}$ the inverse of the covariance matrix. With $D = 1$ and $\boldsymbol{\Sigma} = \sigma^2$ the formula is the ordinary normal PDF.

3. **Example:** $D = 2$, mean $(0, 0)$, $\boldsymbol{\Sigma}$ with diagonal 1, 4 and zeros elsewhere (variances 1 and 4, no covariance). Then $|\boldsymbol{\Sigma}| = 4$ and $\boldsymbol{\Sigma}^{-1}$ has diagonal 1, 1/4. At $\mathbf{x} = (1, 2)$ the quadratic form is $1^2/1 + 2^2/4 = 2$:

$$N = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{4}} e^{-1} = 0.0293$$

Points with the same density lie on an ellipse around the mean (the book's Figure 6.8b). Variances stretch the ellipse along the axes; a non-zero covariance tilts it. Figure 5 draws both cases with the example's variances 1 and 4: without covariance the ellipses stand upright, twice as tall as wide; with a covariance of 1.6 they tilt.

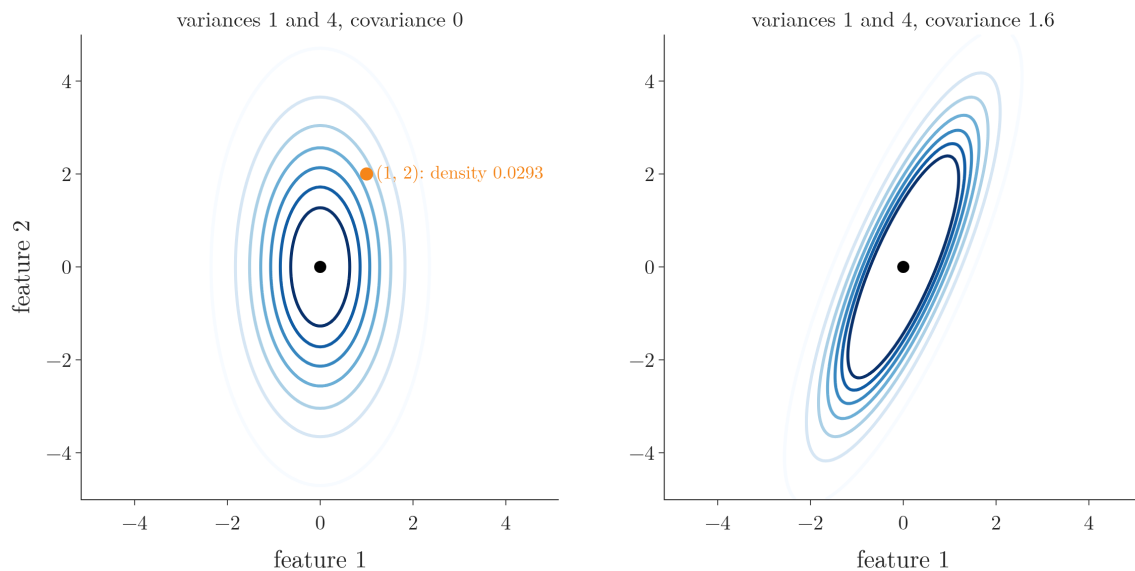


Figure 5: Contours of two-dimensional normal components with variances 1 and 4. Left: no covariance, ellipses along the axes; the example point (1, 2) has density 0.0293. Right: covariance 1.6, tilted ellipses.

Figure 9 (right, Section 9) shows tilted ellipses fitted to real flowers.

5.2 The mixture in D dimensions

Key point: Same formula as in 1D, with vectors and matrices.

The mixture is $p(\mathbf{x}) = \sum_k \pi_k N(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, with one mean vector and one covariance matrix per component. Everything in the rest of this Note works the same way in any number of dimensions.

6. Responsibilities

Key point: The responsibility r_{nk} is the probability that component k produced observation x_n . The responsibility is the posterior of Bayes' theorem: weight times density, divided by the mixture density.

6.1 The formula

Key point: $r_{nk} = \pi_k N(x_n \mid \mu_k, \sigma_k^2) / \sum_j \pi_j N(x_n \mid \mu_j, \sigma_j^2)$.

Having seen a value x_n , which component produced it? **Bayes' theorem** (G-269; see the [Bayes' theorem Note](#)) answers this with the prior $P(z = k) = \pi_k$, the likelihood $N(x_n \mid \mu_k, \sigma_k^2)$ and the evidence $p(x_n)$.

1. **In words:** the **responsibility** (G-1687) of component k for point n is that component's share of the mixture density at x_n .
2. **Formula** (MML equations 11.17 and 11.72):

$$r_{nk} = P(z_n = k \mid x_n) = \frac{\pi_k N(x_n \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^K \pi_j N(x_n \mid \mu_j, \sigma_j^2)}$$

3. **Example:** the book's running example (Section 11.2, Example 11.1) has seven points $-3, -2.5, -1, 0, 2, 4, 5$ and a starting mixture $N(-4, 1)$, $N(0, 0.2)$, $N(8, 3)$ with weights $1/3$ each. For $x_3 = -1$ the weighted densities are

$$\frac{1}{3}N(-1 \mid -4, 1) = 0.00148, \quad \frac{1}{3}N(-1 \mid 0, 0.2) = 0.02441, \quad \frac{1}{3}N(-1 \mid 8, 3) \approx 0$$

Their sum is 0.02589, so $r_{31} = 0.00148/0.02589 = 0.057$, $r_{32} = 0.943$ and $r_{33} = 0$.

For each point the responsibilities add up to 1, because each numerator is one term of the denominator.

6.2 Soft assignment

Key point: A point is not assigned to one component; it is shared among them in proportion to the responsibilities.

All seven points give this table (the book's equation 11.19; the Notebook reproduces it):

x_n	-3	-2.5	-1	0	2	4	5	total N_k
r_{n1}	1.000	1.000	0.057	0.000	0.000	0.000	0.000	2.057
r_{n2}	0.000	0.000	0.943	1.000	0.066	0.000	0.000	2.009
r_{n3}	0.000	0.000	0.000	0.000	0.934	1.000	1.000	2.934

Each column of numbers is one point's **soft assignment** (G-1826). The last column, $N_k = \sum_n r_{nk}$, is the **total responsibility** (G-1992) of component k : about how many points it accounts for. The three totals add up to 7, the number of points.

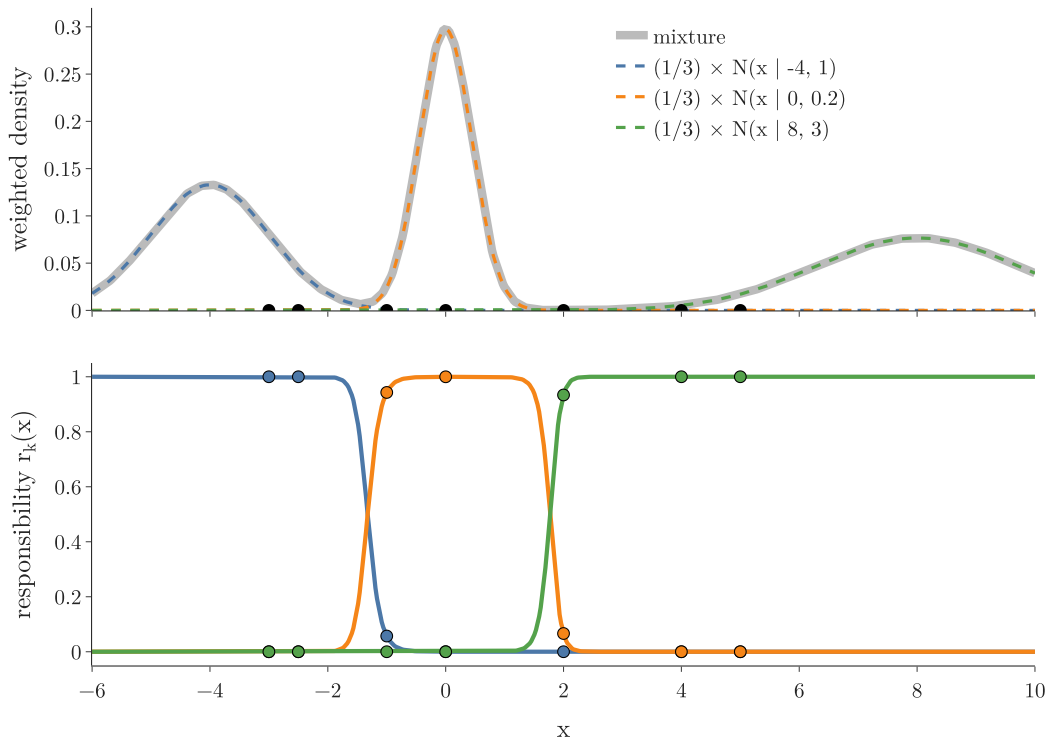


Figure 6: Top: the starting mixture of the book's example and its seven points. Bottom: the responsibility of each component across x , with the seven points marked

Figure 6 (bottom) draws the responsibilities for every x . Near a component's centre its responsibility is close to 1; between two components it changes from one to the other, and there a point is shared, as $x = -1$ is shared 0.057 to 0.943.

7. Why maximum likelihood has no closed form

Key point: The log of a mixture is the log of a sum, which does not split. Setting the derivatives to 0 gives equations for the parameters that contain the responsibilities, and the responsibilities depend on the parameters.

7.1 The log-likelihood

Key point: $\ell(\theta) = \sum_n \log \sum_k \pi_k N(x_n | \mu_k, \sigma_k^2)$: a sum of logs of sums.

For **i.i.d.** (G-933) data the likelihood is a product over the points (the [maximum likelihood estimation Note](#)), and each factor is now a mixture density.

1. **In words:** for each point, add up its weighted component densities, take the log, then add the logs over all points.

2. **Formula** (MML equation 11.10):

$$\ell(\theta) = \sum_{n=1}^N \log p(x_n | \theta) = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k N(x_n | \mu_k, \sigma_k^2)$$

3. **Example:** for the seven points and the starting mixture, $\ell = -28.3$ (the book's Example 11.5; the Notebook gets -28.33).

For one normal curve, the log went straight onto the exponential and left a simple sum of squares (the [MLE for common distributions Note](#), Section 4.2). Here the log sits on a sum over k , and $\log(a + b)$ cannot be split into $\log a + \log b$. The book (Section 11.2, remark after equation 11.10) names this as the reason no closed-form solution exists.

7.2 Setting the derivative for a mean to 0

Key point: The condition “slope = 0” for μ_k says that μ_k is the responsibility-weighted average of the data.

We follow the book's proof of Theorem 11.1 in one dimension.

1. **In words:** differentiate ℓ with respect to one mean, μ_k , set the result to 0, and solve for μ_k .
2. **Formula:** by the chain rule, the derivative of $\log p(x_n)$ is $1/p(x_n)$ times the derivative of $p(x_n)$. Only the k -th term of the sum contains μ_k , and the derivative of a normal density with respect to its mean is the density times $(x_n - \mu_k)/\sigma_k^2$. So

$$\frac{\partial \ell}{\partial \mu_k} = \sum_{n=1}^N \underbrace{\frac{\pi_k N(x_n | \mu_k, \sigma_k^2)}{\sum_j \pi_j N(x_n | \mu_j, \sigma_j^2)}}_{r_{nk}} \cdot \frac{x_n - \mu_k}{\sigma_k^2} = \frac{1}{\sigma_k^2} \sum_{n=1}^N r_{nk} (x_n - \mu_k)$$

The fraction is exactly the responsibility. Setting the sum to 0 and solving:

$$\mu_k = \frac{\sum_n r_{nk} x_n}{\sum_n r_{nk}} = \frac{1}{N_k} \sum_{n=1}^N r_{nk} x_n$$

3. **Example:** with the responsibilities of Section 6.2, component 1 gets

$$\mu_1 = \frac{1.000 \times (-3) + 1.000 \times (-2.5) + 0.057 \times (-1) + 0.0002 \times 0}{2.057} = -2.70$$

The book's Example 11.3 also moves μ_1 from -4 to -2.7 .

In the same way (the book's Theorems 11.2 and 11.3) the other two conditions are:

- **variance:** $\sigma_k^2 = \sum_n r_{nk} (x_n - \mu_k)^2 / N_k$, the responsibility-weighted variance (the book's equation 11.30);
- **weight:** $\pi_k = N_k / N$, the share of the total responsibility. The weights must add up to 1, so the book derives this with a Lagrange multiplier (equations 11.43 to 11.49; see the [Lagrange multipliers Note](#)).

With all r_{nk} equal to 1 for one component, these are the mean, the MLE variance and the share of points of that component's data, as for one normal curve.

7.3 The circular dependence

Key point: To compute μ_k we need the responsibilities; to compute the responsibilities we need μ_k . The equations do not give the answer in one step.

The formula for μ_k looks like a solution, but the r_{nk} on its right-hand side are computed from all the means, variances and weights (the book's remark after Theorem 11.1). Once μ_1 moves from -4 to -2.70 , the responsibilities change: point -1 now belongs 0.56 to component 1 instead of 0.057. The weighted average with these new responsibilities is -2.34 , not -2.70 , so the new μ_1 no longer satisfies its own equation (Notebook, Section 4).

Figure 7 runs these steps. Watch the bar of point -1 in the bottom panel: it is small while $\mu_1 = -4$, grows to 0.56 once μ_1 moves to -2.70 , and with that bigger share pulls the next weighted mean to -2.34 .

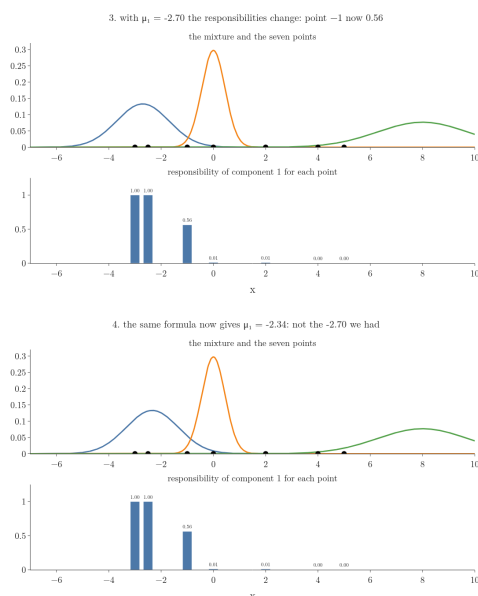


Figure 7: The circular dependence on the seven points. 1: the starting mixture. 2: the slope-zero formula moves μ_1 from -4 to -2.70 . 3: with the new μ_1 the responsibilities change; point -1 now belongs 0.56 to component 1. 4: the same formula now gives -2.34 .

So the three conditions form a set of equations that depend on each other, and MML (§11.2) states that they cannot be solved in closed form. They can be used in turns instead: compute responsibilities, update the parameters, recompute the responsibilities, and so on. Taking turns in this way is the EM algorithm of the [expectation maximization Note](#).

8. When maximum likelihood breaks

Key point: If one component sits exactly on one observation and its variance shrinks to 0, the likelihood grows without limit. Maximum likelihood then prefers a useless spike.

MML (§11.5) warns that maximum likelihood for a mixture can overfit badly in exactly this way. The derivation is the one-point case of the [maximum likelihood estimation Note](#) (Section 4.2, Extra):

1. **In words:** put the mean of component 1 on an observation x_1 . The density of x_1 then contains the term $\pi_1/(\sigma_1\sqrt{2\pi})$, which grows without limit as $\sigma_1 \rightarrow 0$. Every other point keeps a positive density from the remaining components, so the whole likelihood grows without limit.

2. **Formula:**

$$p(x_1 | \theta) \geq \frac{\pi_1}{\sigma_1\sqrt{2\pi}} \rightarrow \infty \quad \text{as} \quad \sigma_1 \rightarrow 0$$

3. **Example:** the seven points, with component 1 placed at -3 , components 2 and 3 fixed at $N(0, 1)$ and $N(4, 1)$, and equal weights. The Notebook (Section 5) gives $\ell = -17.25$ for $\sigma_1 = 0.1$, -14.95 for $\sigma_1 = 0.01$ and -12.65 for $\sigma_1 = 0.001$. Each tenfold shrink adds $\log 10 = 2.30$, the growth of $\log(1/\sigma_1)$ in the formula, and it never stops.

Figure 8 shows the collapse. With $\sigma_1 = 0.1$ the mixture already has a tall, narrow spike on the point -3 , and every tenfold shrink of σ_1 raises the log-likelihood by the same 2.30.

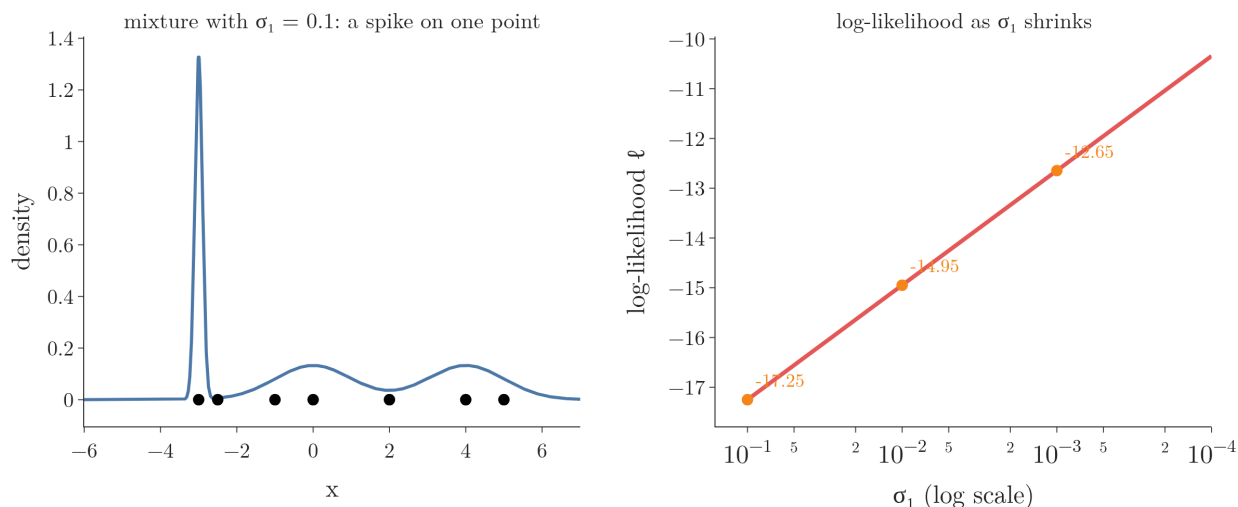


Figure 8: Collapse of maximum likelihood on the seven points. Left: the mixture with component 1 on the point -3 and $\sigma_1 = 0.1$, a spike on one point. Right: the log-likelihood as σ_1 shrinks from 0.1 to 0.0001; it rises in a straight line on the log scale, without limit.

Such a spike describes one point, not a cluster. In practice libraries keep the variances away from 0: scikit-learn's `GaussianMixture` adds a small `reg_covar` (default 10^{-6}) to the diagonal of every covariance matrix, which its documentation says keeps the covariance matrices positive.

9. Mixtures, KDE and k-means

Key point: A GMM is a density estimate with a few learned bumps (KDE has one bump per point), and a soft clustering with elliptical clusters (k-means is hard and uses distance to the centre).

9.1 Compared with KDE

Key point: KDE places one kernel on every observation with a fixed width; a GMM places K components and learns their centres, widths and weights.

The [density estimation Note](#) built a **kernel density estimate** (KDE, G-1005) by placing one small normal curve on every observation and averaging. A GMM also adds normal curves, but only K of them, with every centre, width and weight fitted to the data. The density estimation Note already placed it between parametric and non-parametric methods.

	KDE	GMM
Number of bumps	one per observation (n)	K , chosen by us
Bump centres	the observations	learned (μ_k)
Bump widths	one bandwidth for all	learned per component (σ_k^2 or Σ_k)
Bump weights	all equal, $1/n$	learned (π_k)
Also gives clusters	no	yes, via the responsibilities

9.2 Compared with k-means

Key point: k-means gives each point to its nearest centre; a GMM shares each point by responsibilities and models each cluster's shape with a covariance matrix.

k-means (G-996; the [k-means intuition Note](#)) assigns each point to its nearest centroid and moves each centroid to the mean of its points. MML (§11.5) relates the two: treat the GMM means as cluster centres and ignore the covariances (set them to the identity), and the result is k-means; k-means makes a hard assignment, a GMM a soft one through the responsibilities.

Figure 9 compares the two on the Iris dataset (scikit-learn's `load_iris`): 150 flowers, each with four features (sepal length, sepal width, petal length, petal width) and a known species. Both methods see only the four features; the species is used afterwards to score them. The **adjusted Rand index** (ARI; G-175) measures how well a clustering matches the true groups: 1 for a perfect match, about 0 for random labels (scikit-learn's `adjusted_rand_score` documentation).

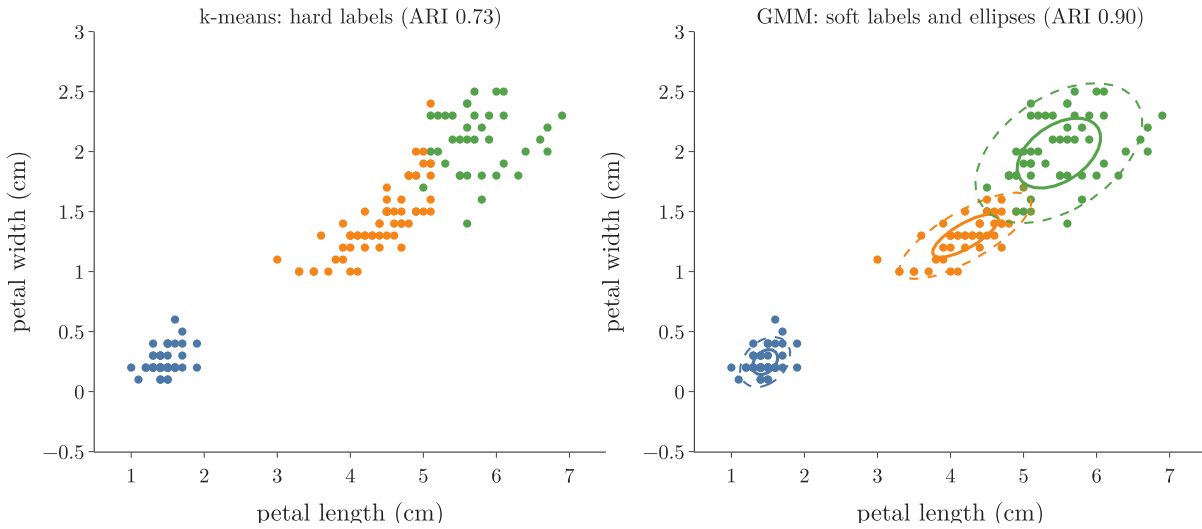


Figure 9: Iris, petal length against petal width. Left: k-means labels. Right: GMM with full covariance matrices; colours mix by responsibility, ellipses show 1 and 2 standard deviations

Method (all four features)	ARI
k-means	0.73
GMM, full covariance matrices	0.90
GMM, round components (one variance per component, no tilt)	0.73

The two species on the upper right overlap, and their clouds are tilted ellipses: longer petals come with wider petals. k-means draws its boundary by distance to the centres, as if every cluster were a circle.

To test whether the covariance matrices make the difference, the Notebook (Section 6) refits the GMM with only that changed: round components, each with a single variance and no tilt. The ARI drops to 0.73, exactly k-means's score. So on Iris the advantage comes from modelling each cluster's stretched, tilted shape.

Python: scikit-learn fits a GMM with the EM algorithm.

```

from sklearn.mixture import GaussianMixture

gmm = GaussianMixture(n_components=3,
                      covariance_type="full",
                      random_state=0).fit(X)

gmm.weights_      # the mixture weights  $\pi_k$ 
gmm.means_        # one mean vector per component
gmm.covariances_  # one covariance matrix per component
gmm.predict_proba(X)  # responsibilities, one row per observation
gmm.predict(X)    # hard labels: the largest responsibility

```

```
gmm.score(X)          # average log-likelihood per point  
covariance_type sets the shape of the components: "full" (any ellipse), "diag"  
(ellipses along the axes), "spherical" (circles) or "tied" (one shared ellipse).
```

10. Summary

Idea	Formula	Seven-point example
Mixture density	$p(x) = \sum_k \pi_k N(x \mid \mu_k, \sigma_k^2)$	start: $N(-4, 1)$, $N(0, 0.2)$, $N(8, 3)$, weights $1/3$
Weights	$\pi_k \geq 0, \sum_k \pi_k = 1$	$1/3 + 1/3 + 1/3 = 1$
Responsibility	$r_{nk} = \pi_k N(x_n \mid \mu_k, \sigma_k^2) / p(x_n)$	$r_{31} = 0.057, r_{32} = 0.943$
Total responsibility	$N_k = \sum_n r_{nk}$	2.057, 2.009, 2.934
Log-likelihood	$\sum_n \log \sum_k \pi_k N(x_n \mid \mu_k, \sigma_k^2)$	-28.3
Slope = 0 for μ_k	$\mu_k = \sum_n r_{nk} x_n / N_k$	$\mu_1 = -2.70$

- A GMM is a weighted sum of normal curves; it can describe data with several peaks that one normal cannot.
- Data from a mixture is generated in two steps: pick a component with probability π_k , then draw from it. The component is a latent variable.
- In D dimensions each component is a multivariate normal with a mean vector and a covariance matrix; its contours are ellipses.
- Responsibilities are Bayes posteriors: each point is shared among the components.
- The log of a sum does not split; the slope-zero conditions contain responsibilities that depend on the parameters, so there is no closed form.
- A component collapsing onto one point sends the likelihood to infinity; libraries add a small `reg_covar`.
- Compared with KDE, a GMM uses few learned bumps; compared with k-means, it assigns softly and fits elliptical clusters (on Iris, ARI 0.90 against 0.73).

11. Sources

Built from

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Free PDF at mml-book.github.io. §6.5 (multivariate Gaussian, eq. 6.63), Ch. 11 intro, §11.1 (GMM, eqs. 11.3–11.5), §11.2 (likelihood eq. 11.10, responsibilities eqs. 11.17 and 11.19, Theorems 11.1–11.3, Examples 11.1–11.5), §11.4.1–11.4.3 (generative process, latent variable, responsibilities as posteriors), §11.5 (collapse of maximum likelihood; relation to k-means).

Other references

- scikit-learn documentation: `GaussianMixture` (`reg_covar`, `covariance_type`) and `adjusted_rand_score`.
- Old Faithful waiting times: seaborn’s `geyser` dataset (272 eruptions), saved in `data/old_faithful.csv`.
- Iris: Fisher’s Iris dataset as shipped with scikit-learn (`sklearn.datasets.load_iris`).

12. Key terms

Term	Meaning
Gaussian mixture model (GMM)	A density built as a weighted sum of K normal densities, $\sum_k \pi_k N(x \mid \mu_k, \sigma_k^2)$
Component	One of the normal densities in a mixture
Mixture weight π_k	The share of the mixture given to component k ; weights are non-negative and add up to 1
Convex combination	A weighted sum with non-negative weights that add up to 1
Generative process	A step-by-step recipe that produces data from a model
Latent variable	A variable in a model that is never observed, such as the component that produced a point
Multivariate normal distribution	The normal distribution of a vector, set by a mean vector and a covariance matrix
Responsibility r_{nk}	The posterior probability that component k produced point n
Soft assignment	Sharing a point among clusters by probabilities instead of giving it to one
Total responsibility N_k	The sum of the responsibilities of component k over all points
Adjusted Rand index (ARI)	A score for how well a clustering matches the true groups: 1 for identical, about 0 for random
<code>reg_covar</code>	scikit-learn's small number added to the covariance diagonals so no component collapses